



TALK-TO-DATA (T2D)

Phase 2 Annexes: Data & Semantic readiness
assessment

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1. How to use this annex pack

This annex pack contains detailed tools and templates supporting the Phase 1 Framing Guide. It is intended for facilitation, documentation and project delivery. The main guide explains the framing logic; this annex provides the reusable working material.

2. Activity question banks

2.1. Validated source inventory

Area	Questions
Source identification	Which systems, tables, marts, dashboards, BI models, APIs or files are believed to be relevant to the priority use case?
Trusted status	Based on current business and platform knowledge, which assets are considered certified, trusted, experimental, deprecated or no longer maintained?
Ownership	Who is believed to own or maintain each relevant source, table, mart, dashboard or API?
Current usage	Which assets are currently used, formally or informally, to answer the priority business questions?
Accessibility	Are these assets expected to be accessible for discovery, POC and production, or are there known constraints?
Refresh	What is the understood refresh frequency and expected latency of each source?
Suitability	Which assets appear suitable for direct querying, semantic-layer access, reference only or exclusion from MVP?
Known caveats	What known caveats, limitations or concerns should be captured before deeper validation?

2.2. Question-to-data mapping

Area	Questions
Question coverage	Which priority questions can be mapped to candidate data assets?
Metric mapping	Which metrics are needed for each question?
Dimension mapping	Which dimensions are needed for slicing, filtering or comparison?
Source mapping	Which source-of-truth asset should be used for each question?
Initial answerability	Does the question appear answerable, partially answerable or unknown?
Validation need	What must be validated before confirming answerability?
MVP decision	Should the question be included, restricted, deferred or excluded?

2.3. Data model review

Area	Questions
Entities	What are the main entities: customer, account, transaction, product, order, contract, event?
Grain	What is the grain of each fact table or mart?
Keys	What are the primary, foreign and business keys?
Join paths	Which joins are approved and safe?
Join risks	Are there many-to-many joins, duplicate keys or ambiguous relationships?
Time logic	Which dates should be used: order date, invoice date, payment date, snapshot date?
Aggregation	At what level can data be safely aggregated?
Reusability	Can the model support multiple priority questions or only narrow cases?

2.4. Metric discovery

Area	Questions
Metric definition	What is the approved definition for each KPI?
Calculation logic	Is the formula documented in SQL, BI, dbt, semantic layer or analyst logic?
Standard filters	What exclusions apply by default?
Dimensions	Which dimensions are approved for slicing the metric?
Inconsistencies	Do teams define the metric differently?
Ownership	Who owns and approves the metric definition?
Source of truth	Which dashboard, mart, finance pack or semantic model is certified?
Ambiguity	Which business terms require clarification before answering?

2.5. Data quality assessment

Area	Questions
Completeness	Are required fields populated sufficiently?
Freshness	Is data updated frequently enough for the use case?
Duplicates	Are there duplicate customers, transactions, orders or events?
Reconciliation	Does the data reconcile to trusted reports or finance packs?
Nulls	Are there critical null values in metrics, keys or dimensions?
Timeliness	Are there delays, backfills or late-arriving data?
Known issues	What issues are already known by data owners or analysts?
Impact	Which priority questions are affected by quality issues?

2.6. Metadata assessment

Area	Questions
Catalogue	Are relevant assets registered in a data catalogue?
Ownership	Are business and technical owners documented?
Lineage	Can key metrics be traced from report to model, table and source?
Glossary	Are business terms and synonyms documented?
Documentation	Are table, column and metric descriptions usable?
Freshness metadata	Is refresh information visible?
Certification	Can users distinguish trusted from non-trusted assets?
T2D usability	Is metadata structured enough for retrieval by the T2D system?

2.7. Access review

Area	Questions
User groups	Who are the target users for MVP?
Authentication	How are users authenticated?
Authorisation	Should T2D inherit existing permissions or apply a new policy layer?
Row-level security	Are users restricted by region, account, customer, business unit or portfolio?
Column-level security	Which fields must be masked, hidden or restricted?
Sensitive data	Which datasets contain PII, financial, confidential or regulated data?
Aggregation risk	Could users infer restricted information through small cohorts or repeated queries?
Output controls	Can users see SQL, export results or access row-level detail?
Logging	What must be logged for audit and compliance?

2.8. Gap analysis

Area	Questions
Answerability	Which priority questions are answerable, partially answerable or not yet answerable?
Data gaps	What data is missing, inaccessible or not trusted?
Semantic gaps	Which definitions, filters, joins or terms are unclear?
Quality gaps	Which quality issues block or limit answers?
Metadata gaps	What ownership, lineage or documentation is missing?
Access gaps	Which permissions or sensitivity decisions are unresolved?
Remediation	What must be fixed, accepted, restricted or deferred?
MVP decision	Which questions should be included, restricted or excluded from MVP?

3. Output templates

These columns are illustrative examples only. They are intended to show the expected structure and level of detail.

3.1. Validated source inventory

Asset	Type	System	Owner	Source of truth status	Trusted status	Refresh	Access status	MVP relevance	Supported question	Known caveats / comments
finance_revenue_mart	Mart	Data warehouse	Finance Analytics	Authoritative for revenue	Certified	Daily	Available	High	Revenue by region	Official source for revenue reporting
Sales pipeline dashboard	BI dashboard	Power BI / Tableau	Sales Ops	Reference only	Trusted	Daily	Available	Medium	Pipeline conversion, sales validation	Useful for metric validation, not direct querying
crm_opportunity_raw	Raw table	CRM replica	CRM Platform	Not authoritative for MVP	Experimental	Near real-time	Restricted	Low	Opportunity-level exploration	Raw object; not suitable for MVP without curation

3.2. Question-to-data mapping

Priority question	Metric(s)	Dimensions	Candidate source	Initial status	What needs validation	MVP decision
What was revenue last month by region?	Net revenue	Month, region	finance_revenue_mart	Likely answerable	Confirm region hierarchy and revenue definition	Include
Which customers have declining spend?	Customer revenue	Customer, month	Revenue mart + customer master	Partially answerable	Validate customer join and minimum time window	Include if validated
What is churn by segment?	Churn rate	Segment, month	CRM + billing data	Unknown	Confirm churn definition and segment ownership	Pending

Suggested status values:

Status	Meaning
Answerable	Likely answerable with existing trusted data
Partially answerable	Possible, but with caveats or unresolved assumptions
Unknown	Requires further validation
Not answerable	Missing data, definition, access or quality
Out of scope	Not suitable for MVP

3.3. Data model review

Entity / table	Grain	Primary key	Approved joins	Risky joins	Date logic	Known issues	Decision
finance_revenue_mart	Invoice line	invoice_line_id	Customer, product, region	Opportunity join	Invoice date	Month-end adjustments	Use for revenue
customer_master	Customer	customer_id	Revenue, account owner	CRM account ID	Current snapshot	Duplicate legacy IDs	Use with caveat
crm_opportunity	Opportunity	opportunity_id	Account, owner	Revenue mart	Close date	Stage definitions vary	Reference only

3.4. Metric discovery

Metric	Business definition	Calculation logic	Grain	Standard filters	Approved dimensions	Source of truth	Owner
Net revenue	Revenue after discounts, credits and refunds	Sum of net invoice amount	Invoice line	Exclude test accounts, cancelled invoices	Month, region, product, customer segment	Finance revenue mart	Finance
Gross margin	Net revenue less cost of goods sold	Net revenue - COGS	Invoice line	Exclude internal transactions	Month, product, region	Finance pack	Finance
Active customer	Customer with revenue in last 90 days	Count distinct customer ID	Customer-month	Exclude internal/test customers	Segment, region, product	Customer mart	Sales Ops

Ambiguity log

Term	Possible meanings	Default interpretation	Clarification needed?	Owner	Decision
Revenue	Gross, net, recognised, booked	Net revenue	Yes for executive reporting	Finance	Use net revenue by default
Customer	CRM account, billing customer, legal entity	Billing customer	Yes for sales analysis	Sales Ops	Align customer hierarchy
Churn	No spend, cancelled contract, lost account	No spend in last 90 days	Yes	Customer Success	Not MVP until agreed

3.5. Data quality assessment

Dataset / field	Issue	Affected question(s)	Severity	Business impact	Remediation	Owner	MVP decision
customer_master.customer_id	Duplicate legacy customer IDs	Customer trends, churn	High	Over/under-count customers	Apply master customer mapping	Data Engineering	Include if fixed
finance_revenue_mart.region	Region sometimes null	Revenue by region	Medium	Unallocated revenue	Add “unknown region” rule	Finance Analytics	Include with caveat
crm_opportunity.stage	Stage definitions inconsistent	Pipeline conversion	High	Misleading conversion rates	Standardise stage mapping	Sales Ops	Defer

Severity scale:

Rating	Meaning
Low	Minor caveat, does not block MVP
Medium	Needs caveat, restriction or monitoring
High	Blocks answer or requires remediation
Critical	Should exclude from MVP until resolved

3.6. Metadata assessment

Asset / metric	Catalogue entry	Owner	Lineage	Glossary terms	Documentation quality	Certification status	T2D usability	Gap
Net revenue	Yes	Finance	Report → mart → ERP	Yes	Good	Certified	High	None
Customer segment	Partial	Sales Ops	Partial	Partial	Medium	Trusted	Medium	Segment definitions need update
Churn rate	No	Unclear	No	No	Poor	Not certified	Low	Owner and definition required

3.7. Access review

User group	Dataset	Allowed rows	Restricted columns	Sensitive fields	Output restrictions	Export allowed?	Approval needed
Executives	Finance mart	All regions	None	Margin	Aggregated answers only	No	Finance
Regional managers	Revenue mart	Own region	Margin, customer identifiers	Customer name, margin	No row-level detail	No	Regional Finance
Analysts	Curated marts	Approved domains	PII fields masked	Email, phone, salary	SQL visible	Yes, limited	Data Governance

Additional exposure control template:

Exposure type	Allowed	Restricted	Blocked	Notes
Aggregated KPI	Yes	—	—	Default output for most users
Row-level detail	—	Yes	—	Only for analysts with approved access
PII	—	—	Yes	Excluded from MVP
Raw SQL	—	Yes	—	Visible to analysts only
Export	—	Yes	—	Restricted by role and dataset

3.8. Gap analysis

3.8.1. Output: confirmed answerability matrix

Priority question	Final status	Evidence	Blocking issue	Required action	Owner	MVP decision
What was revenue last month by region?	Answerable	Certified finance mart and agreed metric	None	None	Finance Analytics	Include
Which customers have declining spend?	Partially answerable	Revenue data exists, customer master has duplicates	Customer identity issue	Apply master customer mapping	Data Engineering	Include if resolved
What is churn by segment?	Not yet answerable	No agreed churn definition	Semantic disagreement	Agree definition and owner	Sales Ops	Defer

3.8.2. Output: remediation plan

Gap	Category	Impact	Priority	Remediation action	Owner	Dependency	Decision required
Churn definition differs by region	Semantic	Churn questions cannot be answered consistently	High	Agree standard MVP churn definition	Sales Ops	Finance input	Include or defer churn
Margin access unclear	Access	Risk of exposing sensitive commercial data	High	Define role-based margin policy	Finance / Security	CISO approval	Restrict margin answers
Customer duplicates	Data quality	Customer-level trends may be wrong	Medium	Use master customer mapping table	Data Engineering	MDM rules	Include with caveat

4. Activity scorecards

4.1. Validated source inventory

Criterion	Green	Amber	Red
Relevant assets identified	Key systems, marts, BI assets, APIs and files identified for the priority use case	Main assets identified, but some areas still uncertain	Relevant assets unclear or incomplete
Ownership understood	Business and technical owners identified for key assets	Some owners missing or informal	Ownership unclear for critical assets
Trust and authority understood	Trusted status and likely source-of-truth role are clear for key assets	Some overlap, disagreement or uncertainty remains	Trust status unclear and no authoritative source identified
Access and refresh understood	Access route and refresh cadence known for key assets	Some access or refresh details uncertain	Access or refresh constraints unknown
Suitability for MVP	Assets classified as direct query, semantic-layer, reference-only or excluded	Some suitability decisions pending	Cannot determine which assets are usable

4.2. Question-to-data mapping

Criterion	Green	Amber	Red
Priority questions mapped	Most questions mapped	Partial mapping	Few questions mapped
Candidate sources identified	Clear source per question	Multiple possible sources	No clear source
Validation needs clear	Open questions documented	Some gaps unclear	Validation not defined
MVP decision support	Supports include/defer decisions	Some uncertainty	Cannot support scope decision

4.3. Data model review

Criterion	Green	Amber	Red
Grain understood	Grain clearly documented	Some uncertainty	Grain unclear
Keys understood	Keys validated	Some unresolved keys	Keys unreliable
Join paths	Approved joins known	Some risky joins	Joins ambiguous
Date logic	Standard date logic agreed	Multiple date options	No agreed date logic
Aggregation safety	Safe aggregation rules clear	Some caveats	High risk of wrong aggregation

4.4. Metric discovery

Criterion	Green	Amber	Red
Metric definitions	Approved definitions exist	Definitions exist but inconsistent	No agreed definitions
Calculation logic	Logic documented and reusable	Logic partly documented	Logic in people's heads
Filters	Standard filters clear	Some filters unclear	No standard filters
Dimensions	Approved dimensions known	Some dimension risk	Dimensions uncontrolled
Ownership	Metric owners confirmed	Partial ownership	No owner

4.5. Data quality assessment

Criterion	Green	Amber	Red
Completeness	Sufficient for MVP	Some gaps	Major missing data
Freshness	Meets user need	Some delays	Not fresh enough
Reconciliation	Matches trusted reports	Minor differences	Material mismatch
Key reliability	Keys reliable	Some issues	Joins unsafe
Issue transparency	Known issues documented	Partial visibility	Unknown quality state

4.6. Metadata assessment

Criterion	Green	Amber	Red
Catalogue coverage	Key assets catalogued	Partial coverage	Not catalogued
Ownership	Owners clear	Some missing owners	Ownership unclear
Lineage	Lineage available	Partial lineage	No lineage
Glossary	Terms documented	Some terms missing	No glossary
Retrieval readiness	Metadata machine-usable	Some manual work	Not usable for T2D

4.7. Access review

Criterion	Green	Amber	Red
User groups	Clearly defined	Some uncertainty	Undefined
Permissions	Existing model usable	Needs adaptation	No clear access model
Sensitive data	Classified	Partially classified	Unknown sensitivity
Row/column controls	Controls defined	Some gaps	Controls missing
Auditability	Logging needs clear	Partial logging	Audit unclear

4.8. Gap analysis

Criterion	Green	Amber	Red
Answerability known	Clear status for all priority questions	Some unknowns remain	Many questions unresolved
Gaps categorised	Gaps grouped and prioritised	Some gaps unclear	No clear gap view
Owners assigned	Owners confirmed	Partial ownership	No owners
Remediation defined	Clear actions and dates	Some actions vague	No plan
MVP scope decision	Include/defer decisions clear	Some open decisions	MVP scope blocked

5. Overall phase readiness scorecard

Use this at the end of the phase.

Readiness area	Rating	Evidence	Key risk	Action required	Owner
Source readiness	Green / Amber / Red				
Question coverage	Green / Amber / Red				
Data model readiness	Green / Amber / Red				
Metric readiness	Green / Amber / Red				
Data quality readiness	Green / Amber / Red				
Metadata readiness	Green / Amber / Red				
Access readiness	Green / Amber / Red				
Overall MVP readiness	Green / Amber / Red				

Suggested rating logic:

Rating	Meaning
Green	Ready for governed data layer with limited caveats
Amber	Usable for MVP if specific gaps are resolved or accepted
Red	Not ready; material blocker exists
Grey	Not assessed yet

6. Risk, decision, assumption and dependency logs

6.1. Risks

Risk	Impact	Likelihood	Mitigation	Owner	Status
No agreed source of truth for priority metrics	Conflicting answers and low user trust	Medium	Confirm certified sources with business and data owners	Analytics Engineer	Open
Access to required datasets is delayed	Discovery and validation cannot be completed on time	High	Submit access request early and define fallback data assets	Data Architect	In progress
Metric definitions differ across teams	Assistant may generate inconsistent or incorrect answers	High	Create metric decision log and agree MVP definitions	Business SME / Product Owner	Open
Data quality issues are discovered late	MVP scope may need to be reduced	Medium	Run early profiling on high-priority datasets	Data Engineer	Open
Sensitive data exposure is underestimated	Security or compliance approval may be blocked	Medium	Complete access and sensitivity review before prototype build	Security Architect	Open

6.2. Decisions

Decision	Options	Recommended option	Decision owner	Due date	Status
Which data assets are approved for MVP use?	Certified marts only / BI semantic layer / raw tables with controls	Certified marts and approved semantic assets first	Data Owner / Product Owner		Open
What is the default definition of revenue?	Gross revenue / net revenue / recognised revenue	Use finance-approved net revenue definition	Finance Owner		Open
Should raw SQL be visible to users?	All users / analysts only / no users	Analysts only for MVP	Product Owner / Security		Open
Which questions are included in MVP?	All priority questions / only answerable questions / restricted subset	Include answerable and low-risk partially answerable questions	Product Owner		Open
How are unresolved semantic gaps handled?	Block / caveat / ask clarification / defer	Defer high-risk gaps; caveat low-risk gaps	Product Owner / Business SME		Open

6.3. Assumptions

Assumption	Impact if wrong	Validation method	Owner	Status
Certified marts exist for the priority domain	Source inventory may need to include lower-level tables	Confirm with data platform and BI teams	Data Architect	To validate
Existing BI dashboards reflect trusted business logic	Metric contract may be based on incorrect logic	Reconcile dashboard logic with SQL/dbt/semantic layer	Analytics Engineer	To validate
Target users can be mapped to existing access roles	Additional security design may be required	Compare user groups with current BI / warehouse roles	Security Architect	To validate
Priority questions are representative of MVP usage	Evaluation set and governed layer may be incomplete	Validate questions with business SMEs and user interviews	Business Analyst	To validate
Data refresh is sufficient for user expectations	Users may reject answers as outdated	Compare refresh cadence with workflow needs	Data Engineer	To validate

6.4. Dependencies

Dependency	Needed by	Owner	Due date	Status	Impact if delayed
Access to warehouse / BI semantic model	Source inventory, data model review, quality assessment	Data Platform Team		Open	Delays validation and answerability assessment
Priority question list from framing	Question-to-data mapping and metric discovery	Product Owner / Business Analyst		Open	Discovery scope remains unclear
Business SME availability	Metric discovery and ambiguity resolution	Business Lead		Open	Metric decisions may remain unresolved
Data owner input	Source trust status, ownership and quality issues	Data Owners		Open	Source readiness cannot be confirmed
Security / governance input	Access review and exposure model	Security / Governance Lead		Open	Safe MVP scope cannot be agreed
Existing documentation and SQL examples	Metric contract and semantic grounding	BI Team / Analytics Engineering		Open	More reverse engineering require